

Education

University of Toronto

Graduation: May 2025

Honours Bachelor of Science in Computer Science

Toronto, ON

- Art&Science Internship program
- Relevant Courses: Computer Networks, Distributed Systems, Operating System, Database system, Compiler

Technical Skills

Languages: TypeScript, JavaScript, Java, C++, C, C#, SQL, Go

Technologies: Git, Docker, Kubernetes, Next.js, React, SQL Server, Linux, Express.js, ASP.NET, Node.js, PostgreSQL

Concepts: Agile methodologies, CI/CD pipeline, RESTful API, System Design, Data Structures, LLVM, OOP

Cloud Technologies : AWS DynamoDB, AWS LightSail, Kubernetes CKAD (ongoing), Docker

Experience

ServiceNext Inc - University Industry-Partner Project

Sep 2024 – Dec 2024

Full-stack Developer

Toronto, ON

- Implemented responsive front-end pages of shopping cart, Appointment, payment using **React with vite**.
- Developed **RESTful APIs** using **Express.js** and **PostgreSQL** on backend for efficient data management, improving code quality and data consistency.
- Created comprehensive documentation covering installation, feature introduction, and usage guidelines.

Ontario Financing Authority

May 2023 – Oct 2023

Programmer Intern

Toronto, ON

- Developed **windows form application** that connects to **SQL server** database to fetch, display, filter data through stored procedure using **C#** and Visual Studio.
- Developed a Web Application using **ASP.NET MVC** for fetching, searching, and retrieving data from an **SQL Server database**, with functionality to add and delete database instances from front-end input.
- Created stored procedures in **SQL Server** improving query efficiency, for efficient data fetching and filtering. Conducted **unit tests** in Visual Studio for financial computing program, achieving **95%** test coverage.

Projects

BusTub Storage Engine (CMU 15-445 Self-Study) | C++

May 2026

- Built core components of a disk-oriented DBMS in **C++ 20**, including a thread-safe buffer pool manager, **ARC** page replacement policy, asynchronous disk scheduler, and RAIL-based page guards for concurrent page access.
- Implemented a concurrent **B+Tree** index supporting insertion, deletion, point lookup, iterator-based range scan, page split/merge redistribution, and latch crabbing for fine-grained concurrency control.
- Developed vectorized query execution components and query optimizations, including aggregation, join executors, index scan optimization, external **merge sort**, and **hash join** execution in a relational query engine.
- Implemented **MVCC** transaction processing with snapshot isolation and optimistic concurrency control (OCC), including undo-log version chains, tuple reconstruction, serializable validation, and concurrent index maintenance.

MiniC Compiler | C++

April 2025

- Built a full MiniC compiler in **C++** with **LLVM**, implementing grammar parsing (ANTLR4), AST generation, semantic analysis, and LLVM IR code generation.
- Designed and implemented custom LLVM passes to perform register promotion and common subexpression elimination (CSE), improving the runtime performance of the generated code.

Fault-tolerant Key/Value Service | Go

April 2025

- Implemented the full **Raft** consensus algorithm in **Go**, including **leader election**, **log replication**, and **crash recovery** through persistent state, ensuring fault-tolerant replication across distributed nodes.
- Developed a **linearizable** key-value storage system backed by Raft, supporting **fault tolerance**, leader re-election, and client request deduplication under network partitions and node crashes.

Scriptorium : Online Code Execution Platform | NextJs, TypeScript, React, Node.js

Oct 2024

- Built and deployed a responsive full-stack web application using **Next.js**, Prisma, **React**, Tailwind CSS, and RESTful APIs on **AWS Lightsail**. Created a user-friendly interface with color theme switch and seamless front-end validations.
- Developed online code editing with syntax highlighting, real-time execution & Output, and secure sandboxing via **Docker**, supporting 10+ programming languages.
- Implemented user **authentication (JWT)**, profile management, and features for saving, searching, editing, and sharing code templates, including template forking.
- Developed a blog platform with post creation, comments, upvotes/downvotes, reporting functionality, and dynamic interactions with code templates.